



PEP 559

Machine Learning in Quantum Physics

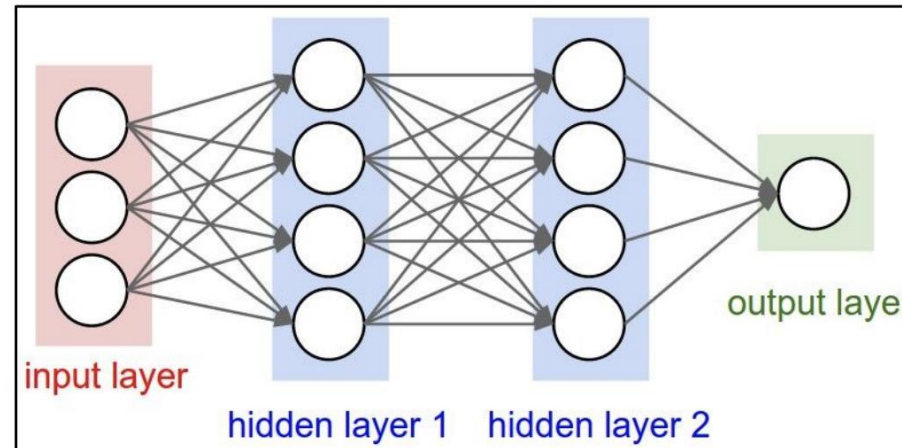
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Recap: MLP and mini-batch SGD

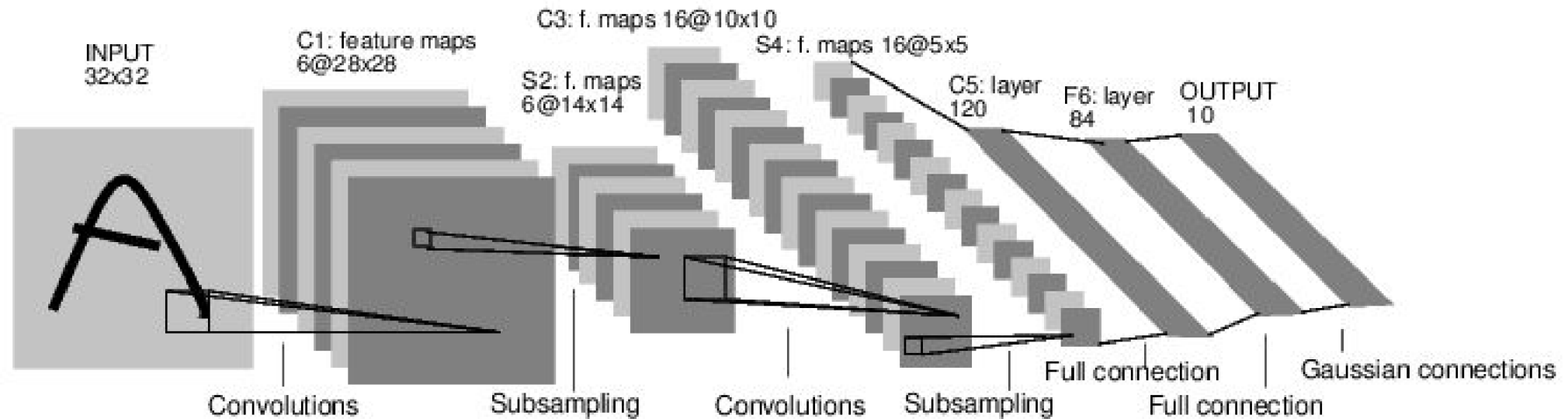
Loop:

1. **Sample** a batch of data
2. **Forward** prop it through the graph, get loss
3. **Backprop** to calculate the gradients
4. **Update** the parameters using the gradient



Convolutional Neural Networks (CNNs)

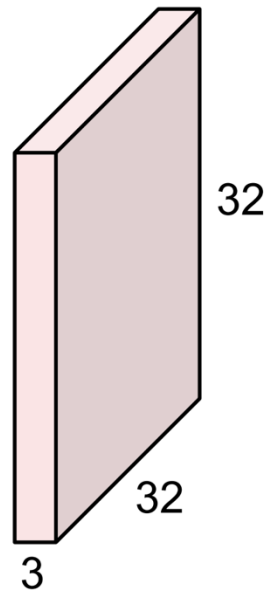
- LeCun Yann et al., 1980
- For classifying handwritten digits from images



Convolution layer: **feature extraction layers**

- 32=height (pixels)
- 32=width (pixels)
- 3 = number of channels – (RGB color)
- For grayscale image, only one intensity value per pixel, and hence figure size=32x32x1 or just 32x32

32x32x3 image



5x5x3 filter



Convolve the filter with the image
i.e. “slide over the image spatially,
computing dot products”

Key ideas

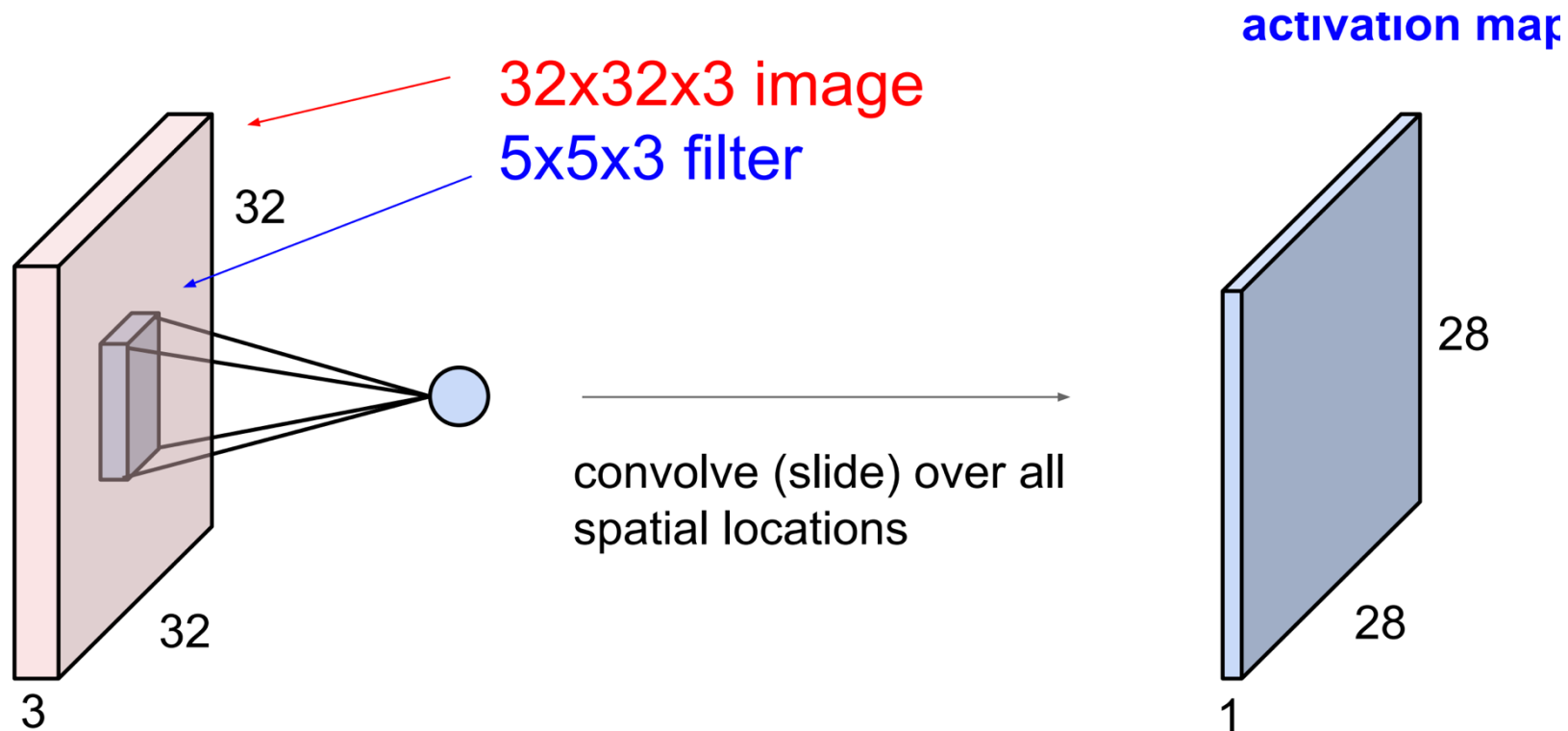
CNN performs very well on image-related tasks, because

- **Sparse connectivity**: A single element in the feature map is connected to only a small patch of pixels --- local pixels are more correlated --- edges, textures, and features usually appear locally
- **Parameter sharing**: The same filter weights are used for different patches of the input images

Replacing a conventional, fully connected MLP with a convolution layer substantially decreases the number of weights (parameters) in the network

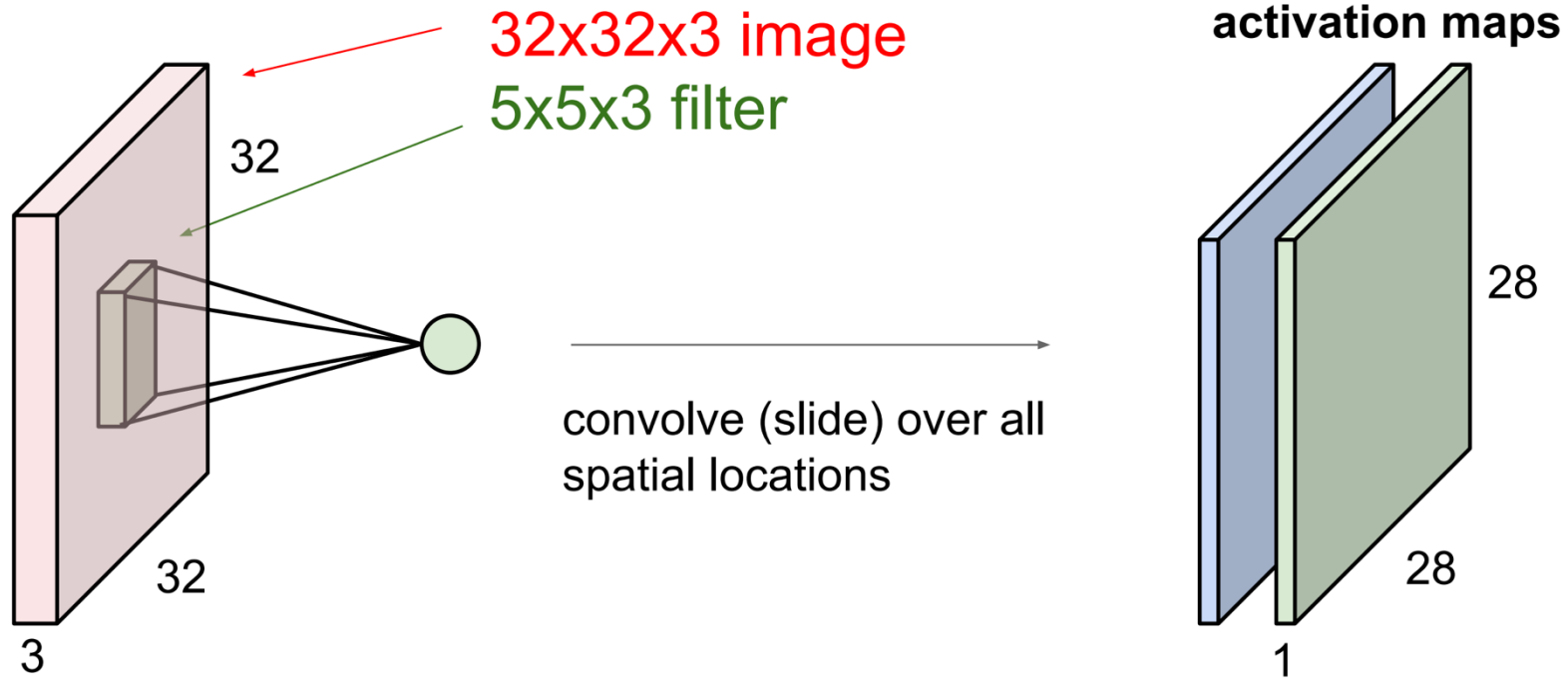
Convolution layer

- After the convolution, each filter generates an activation map



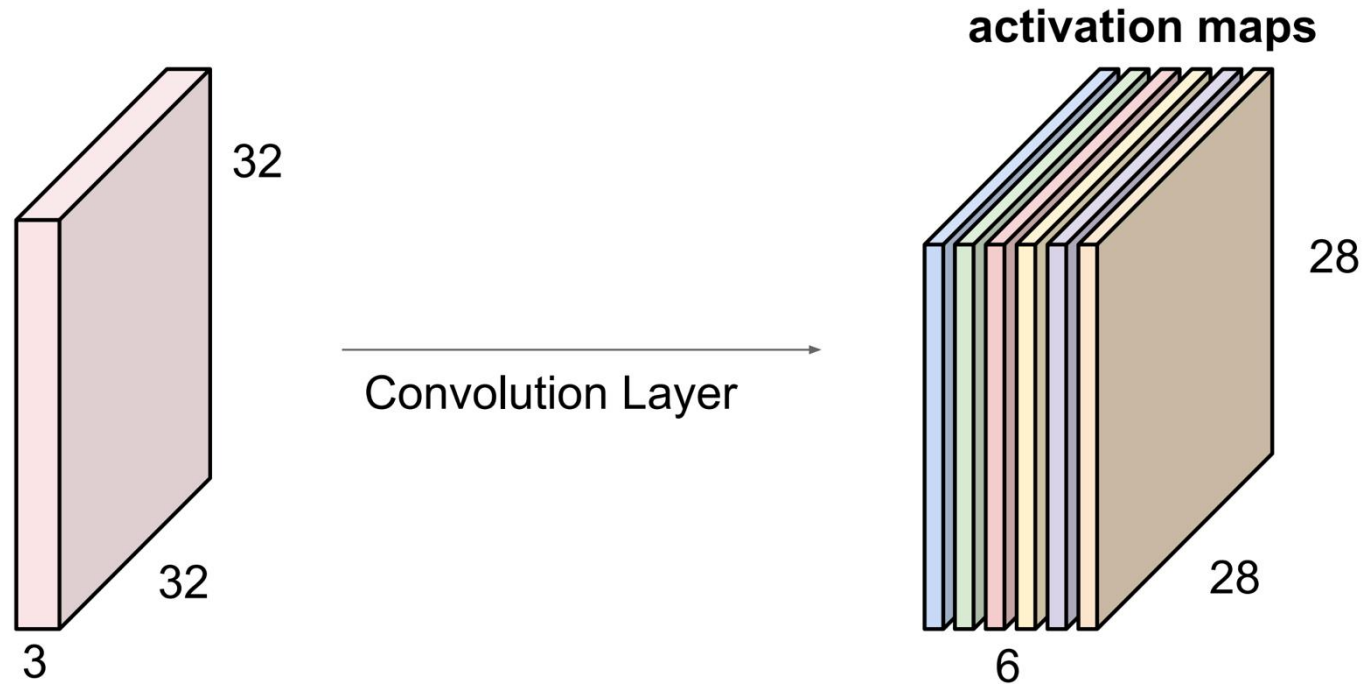
Convolution layer

- A new filter generates a second map



Convolution layer

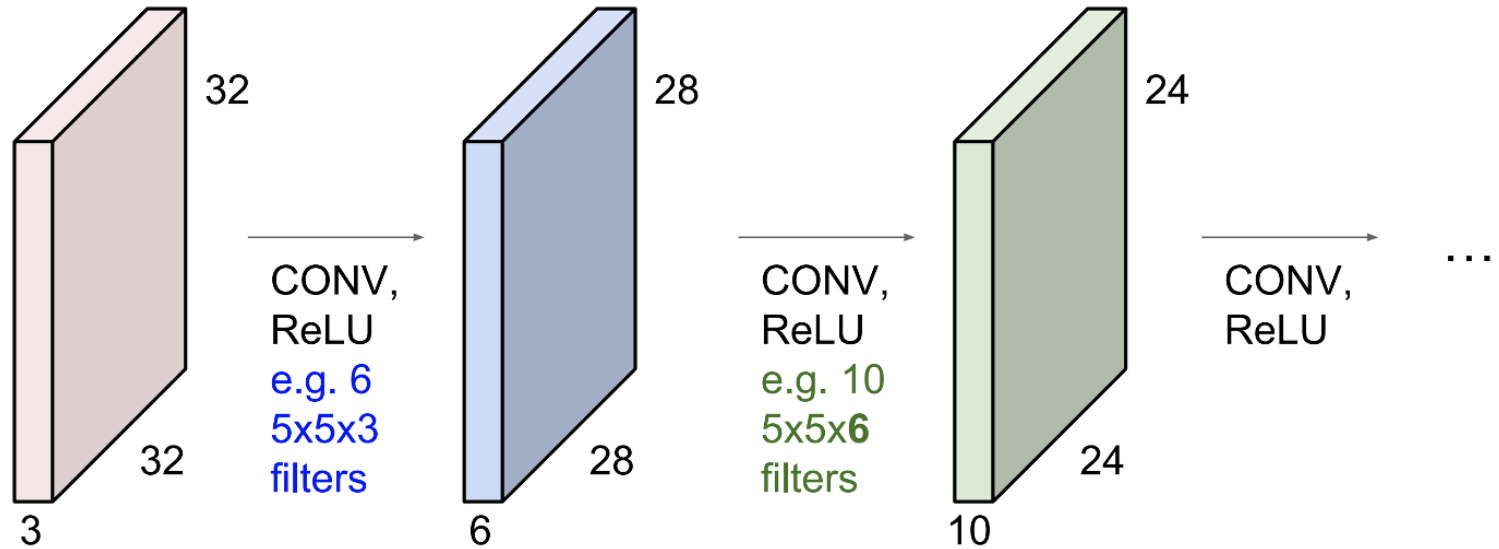
For example, if we had 6 5x5 filters, we'll get 6 separate activation maps:



We stack these up to get a “new image” of size 28x28x6!

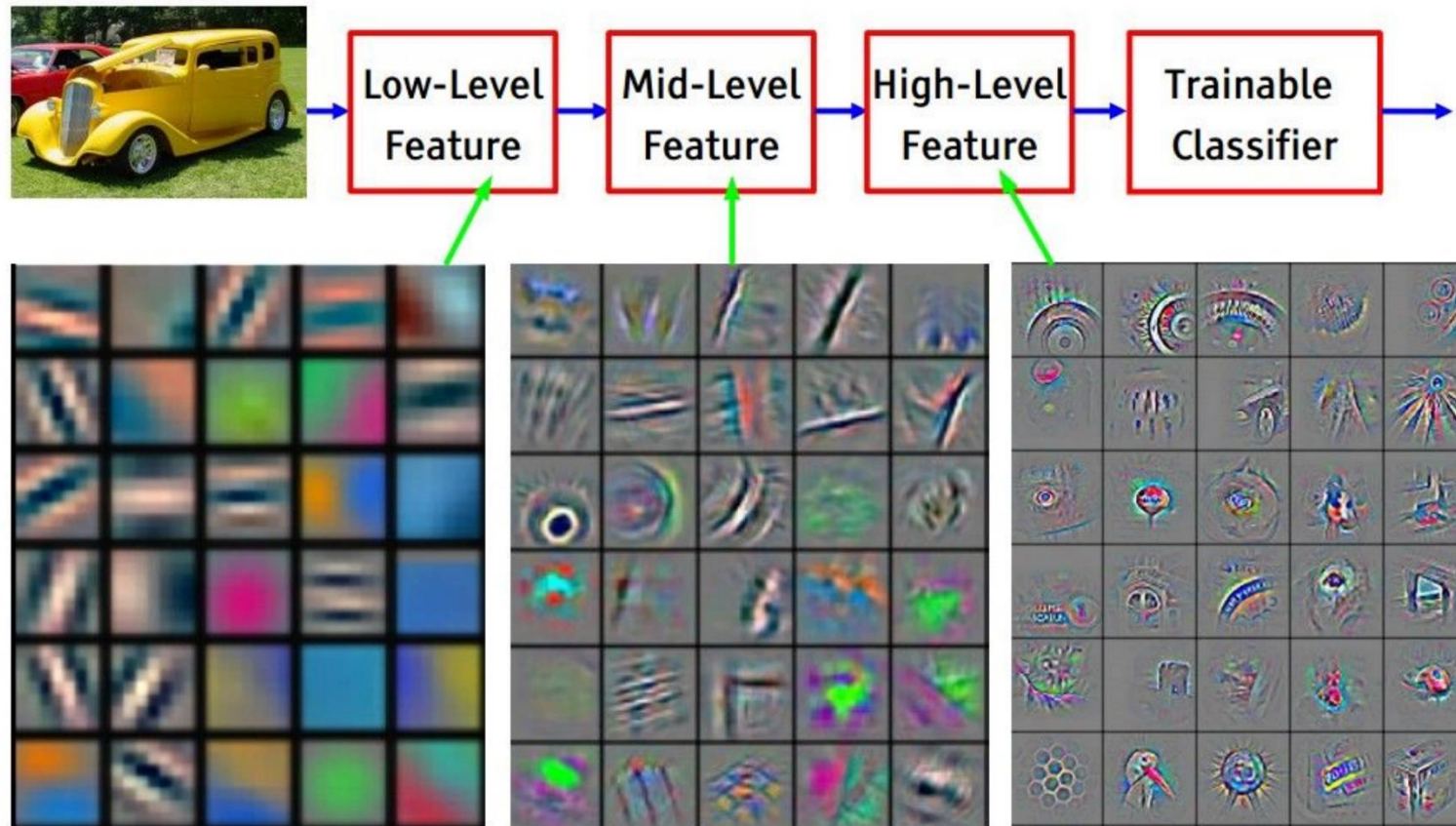
ConvNet architecture

- A sequence of convolutional layers, interspersed with activation functions

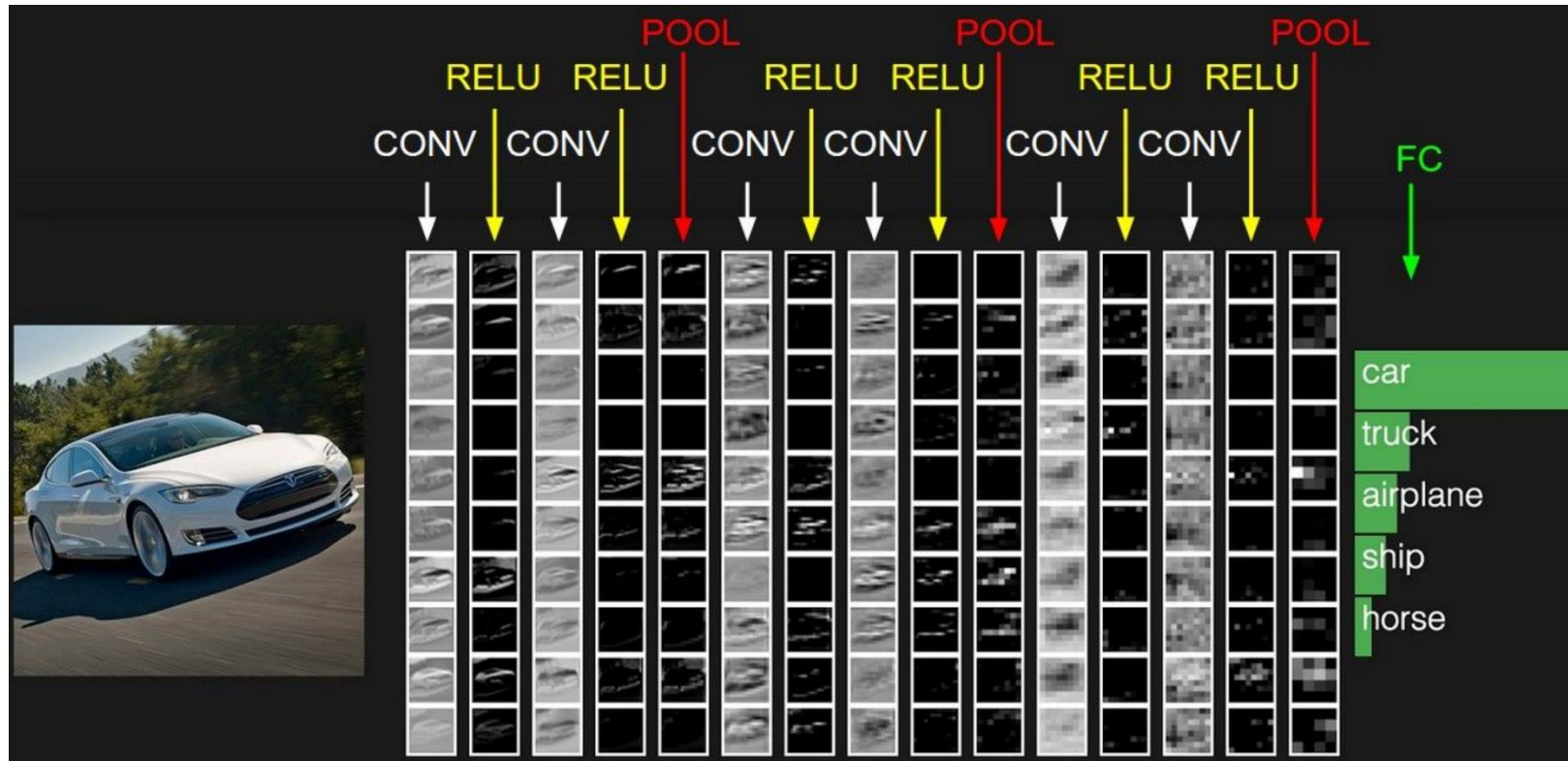


Example: image recognition

- Different kernels extract different levels of features (edge, textures, shape, parts, etc)

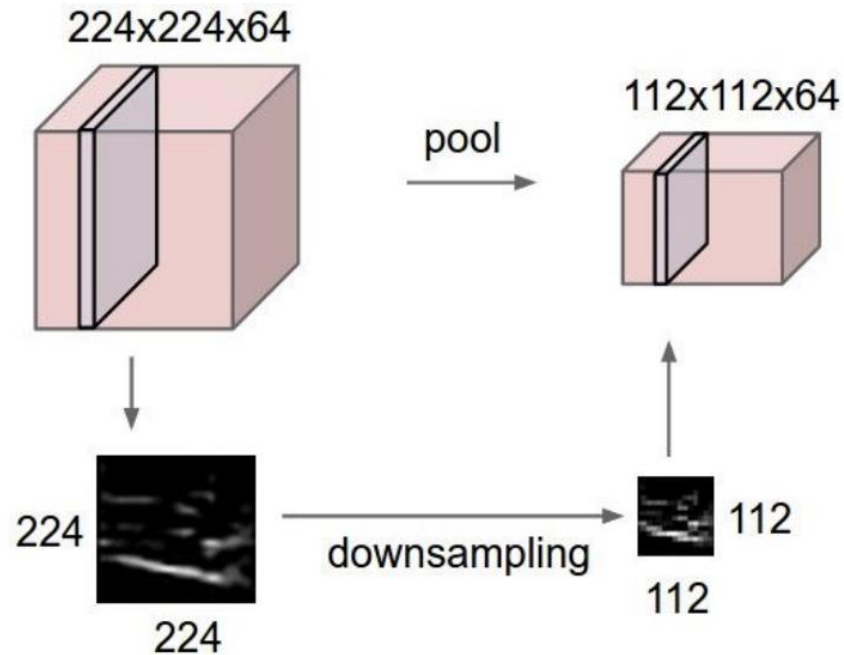


Example: ConvNet architecture for image recognition



Pooling layer

- makes the representations smaller and more manageable
- operates over each activation map independently:



Fully-connected layer

- Contains neurons that connect to the entire input volume, as in ordinary Neural Networks

